

Syrudas AI: A Local-First AI Workspace with Pluggable Model Providers

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In brief

Syrudas AI is a program you install on your own Windows PC. It gives you a chat window for talking to an AI, much like the well-known online assistants, with one difference: nothing you type has to leave your computer.

You choose which AI answers you. It can be one running entirely on your own machine — downloaded once and stored locally — so your conversations never touch the internet at all. Or it can be one of the commercial services, if you have an account and prefer their answers. Either way you use the same window, the same saved history, and the same settings. Switching between them takes a few seconds and changes nothing else about how the program behaves.

Beyond conversation, the AI can be given permission to *do* things: run commands, read and write files in folders you nominate, search the web, and remember facts between conversations. Anything genuinely risky stops and asks you first — every time, showing you exactly what it proposes to do before it does it. It can also read your own documents, such as a folder of notes or reports, and answer questions about them without uploading anything.

The rest of this paper explains how the program is built and why. It is written for readers who want that level of detail. **Key terms**, immediately below, defines the technical vocabulary that recurs throughout; Sections 1 and 2 give the reasoning and the overall shape; the remaining sections take each part in turn, and Section 19 is a candid account of what the system does not yet do well.

Contents

In brief	1
Key terms	2
Abstract	4
1. Motivation	4

2. System overview	5
3. The provider contract	6
4. The normalized interior	7
5. The chat pipeline	8
6. Agent mode	10
7. Persistent memory	12
8. Knowledge: local retrieval	12
9. Deep research	13
10. The writing editor	14
11. The blind arena	14
12. The model cookbook	15
13. Interoperability	15
14. File attachments	16
15. Security and privacy model	16
16. Accessibility and theming	18
17. Testing and assurance	19
18. Packaging and distribution	20
19. Limitations and roadmap	21
20. Conclusion	22

Key terms

Term	What it means here
Model	The AI itself — the component that reads your message and writes a reply.
Provider	A service that runs a model. May be software on your own PC or a company's website.
Local model	A model running on your own hardware, with no internet involved.
Token	The unit models read and write in — roughly three-quarters of a word.
Context window	How much of a conversation a model can consider at once, measured in tokens. Older material falls out when it fills.
Streaming	Showing a reply word by word as it is produced, rather than waiting for all of it.
Agent mode	A setting where the AI may take actions on your behalf, not just reply.
Tool	One specific action the AI can take, such as reading a file or searching the web.
Tool call	The AI's request to use a tool, which the program then carries out.
Approval gate	The prompt asking your permission before a risky tool runs. Nothing risky bypasses it.
Plugin	A small file that teaches the program to talk to a new provider, added without changing the program itself.
MCP	Model Context Protocol — a shared standard letting outside programs offer tools to the AI.
Retrieval	Searching your own documents for passages relevant to a question, and showing them to the model.
Embedding	A numeric fingerprint of a passage of text, used to find passages with similar meaning.
Prompt injection	Text hidden in a web page or document that tries to issue instructions to the AI as though you had written them.
Loopback	The address 127.0.0.1, meaning "this computer only" — unreachable from the network.

Abstract

Syrudas AI is a self-hosted AI workspace for Windows that runs entirely on the user's own machine. It offers streaming chat, an autonomous agent mode with tool use and Model Context Protocol (MCP) support, durable cross-conversation memory, local retrieval over the user's own files, a deep-research pipeline, a blind model-comparison arena, an AI-assisted writing editor, a hardware-aware model cookbook, and an OpenAI-compatible endpoint that lets other tools use the same configured backends. What makes this breadth tractable for a single maintainer is a deliberate architectural bet: **the model is a plugin**. Every backend — local weights or a hosted frontier API — reaches the application through one small contract, and nothing above that contract knows which model is answering. This paper is a from-first-principles account of the system: the thesis, the abstraction that carries it, the normalized data model that makes the abstraction cheap, how each feature falls out of that model as a consequence rather than a bolt-on, and the security posture that a local, single-user tool can honestly promise.

1. Motivation

Using a large language model increasingly means renting a seat in a vertically integrated stack: one vendor's model, behind one vendor's application, with the user's conversations, keys, and files living on one vendor's servers. That arrangement is convenient and, for many, sufficient. It is also fragile in a particular way: it couples the *interface* you rely on to the *model* you happen to prefer this quarter, and it couples both to a third party's continued goodwill about your data. Projects such as PewDiePie's **Odysseus** (2026) showed that a meaningful audience wants the opposite: a workspace you run yourself, where the conversation history, the API keys, and the documents never leave the machine.

Syrudas AI begins from that local-first premise but pushes on one idea harder than most such projects do. Model backends are the fastest-moving, least-durable part of this ecosystem — new weights and new APIs arrive monthly, and today's best choice is next quarter's fallback. A workspace built around any specific model, or any specific provider's SDK, inherits that churn. So the organizing principle here is that **the model is the most replaceable component, and the architecture should treat it that way**. Everything the user actually values — their history, their tools, their documents, the agent runtime, the UI — sits above a thin seam, and the model plugs into that seam.

Five concrete goals follow from this stance:

1. **Any model, one workspace.** Local weights via Ollama or LM Studio, and hosted models via OpenRouter or first-party APIs, share one UI, one conversation store, and one agent runtime.
2. **Local-first and private by construction.** The server binds to `127.0.0.1` and rejects non-loopback requests. There is no account, no telemetry, and no cloud component. Keys are stored locally and masked in every response.
3. **Agentic, but consent-gated.** The model can plan and act — run commands, read and write files, search and read the web, remember facts, search indexed documents — with an explicit, per-action human gate on the dangerous parts.
4. **A hub, not a silo.** The workspace speaks the OpenAI wire dialect in both directions, so external tools can use its configured models as a service.
5. **Shippable to a non-developer.** A single portable executable that unzips and double-clicks, with zero-configuration onboarding when a local model server is already present.

The rest of this paper traces how a single abstraction, enforced everywhere, lets those goals coexist inside a codebase one person can hold in their head.

2. System overview

Syrudas is a two-tier application with a deliberately small surface between the tiers. The **backend** is Python 3.13 on FastAPI: it serves a REST and streaming API, owns a single SQLite database, runs the agent and research loops, and hosts the provider, MCP, and retrieval subsystems. The **frontend** is a React 19 / TypeScript single-page app built with Vite — chat, editor, arena, cookbook, and settings — compiled to static assets that the backend serves. The same compiled bundle runs unmodified in an ordinary browser tab and in the packaged desktop window; nothing about it assumes which.

The packaged desktop application wraps both tiers in a native window (pywebview over the Windows WebView2 runtime), with the FastAPI server running on a background thread of the same process. There is no second process to supervise and no port coordination beyond a single fixed local port; closing the window stops the server, and launching the executable again while it is running simply opens a new window onto the existing instance.

State is intentionally boring, which is a feature rather than an apology. One SQLite file holds conversations, messages, provider instances, MCP registrations, memories, indexed-document chunks and their embeddings, editor documents, arena results, and settings. One workspace folder holds files the agent writes. In the portable build both live directly beside the executable, so copying the folder copies the entire installation and deleting it removes every trace — no registry keys, no scattered application-data directories. Lightweight session state — the open view and conversation, plus the model and theme choices — is kept in the frontend's local storage (persisted across desktop restarts by the WebView2 storage path), so the app reopens exactly where the user left off.

```
+----- SyrudasAI.exe -----+
| pywebview window (WebView2) |
| +-- React SPA -- NDJSON/HTTP --> FastAPI (127.0.0.1:8040) |
| | chat · agent · research · | | +-- Host-guard middleware | | | |
| | knowledge · arena · editor · | | +-- /api/chat · /api/research |
| | cookbook · settings | | +-- /api/documents · /api/knowledge |
| | | | | | +-- /api/arena · /api/cookbook |
| | | | | | +-- /v1 (OpenAI-compatible hub) |
| | | | | | +-- Agent loop -- builtin tools |
| | | | | | | +----- MCP client --> stdio servers |
| | | | | | +-- Provider registry |
| | | | | | | +-- openai_compat (builtin) |
| | | | | | | +-- plugins/*.py (drop-in) |
| | SQLite (syrudas.db) · data/workspace · plugins/ |
+-----+
|
| ^
| Continue / aider / any
| OpenAI-compatible client via /v1
```

3. The provider contract

Everything in the previous section rests on a single class. A provider *type* is a Python class; a provider *instance* is that class configured with user data — a base URL, an API key — in the settings UI and persisted in SQLite. The entire contract is five methods, two of them optional:

```
class ModelProvider(ABC):
    type_id: str # "openai_compat", "anthropic", ...
    display_name: str
    config_fields: list[ConfigField] # declares the settings form to render

    async def list_models(self) -> list[ModelInfo]: ...
    async def chat(self, model, messages, tools=None, params=None)
        -> AsyncIterator[StreamEvent]: ...
    async def embed(self, model, texts) -> list[list[float]]: ... # optional
    async def check(self) -> str: ... # optional
```

Three decisions turn this small surface into the load-bearing wall of the whole system.

A normalized interior. Messages, tool specifications, tool calls, and stream events are defined exactly once, in an OpenAI-flavored shape (Section 4). A

provider adapter's only job is to translate between that shape and its backend's wire format, at the edge. The chat pipeline, the agent loop, the persistence layer, and the UI are all written against the normalized types and never learn which backend produced them. Adding support for a new API is therefore a self-contained translation file, not a change that ripples through the application.

Streaming as the only mode. `chat()` returns an async iterator of stream events; there is no separate "get a completion" method. A non-streaming response is simply a stream that ends quickly. This collapses what is usually two code paths — with two sets of bugs — into one, and it means every consumer, from the chat view to the deep-research synthesizer, handles partial output the same way.

Discovery by drop-in. The registry imports its builtin adapters statically, because a PyInstaller onefile bundle is invisible to `pkgutil`'s directory scanning and dynamic-only imports would never be packaged at all. On top of that, it scans a user-writable `plugins/` folder next to the executable at startup. The consequence is unusual and, for a local tool, delightful: a user can teach the *packaged* application to speak to a brand-new backend by dropping a single Python file beside the exe and restarting — no rebuild, no toolchain.

The single builtin adapter, `openai_compat`, is enough for most of the ecosystem, because Ollama, LM Studio, the llama.cpp server, vLLM, OpenRouter, and OpenAI itself all speak the same dialect. Optional connectors for **Anthropic (Claude)** and **Google (Gemini)** ship as drop-in plugins that translate those non-OpenAI dialects. The `embed()` method is the one place the contract grew a new capability rather than a new backend: providers that implement it become eligible to power the retrieval subsystem (Section 8), while providers that do not simply raise `NotImplementedError` and are omitted from the embedding-model picker.

4. The normalized interior

The abstraction in Section 3 is only cheap because the types it exchanges are few and stable. They live in one module and are shared verbatim by every provider, route, and persistence call:

- `Message` — a role (`system / user / assistant / tool`), text

content, an optional list of tool calls (on assistant messages), and an optional tool-call id (on tool-result messages).

- `ToolSpec` — a tool offered to the model: name, description, and a

JSON-Schema parameter object.

- `ToolCall` — a model's request to invoke a tool: `id`, `name`, `parsed arguments`.
- `StreamEvent` — one event in a provider's output stream. The vocabulary is small and closed: `text_delta`, `tool_call`, `usage`, `error`, `done`.

The event vocabulary is where the "streaming as the only mode" decision pays off structurally. The chat route serializes these events to the client as newline-delimited JSON, but along the way it *interleaves* events the provider never produced — an `approval_required` when the agent needs consent, a `research_status` as the research pipeline works, a `tool_result` after a tool runs. Because the client already consumes one flat event stream, plain chat, agent tool activity, research progress, and consent prompts all arrive over the same channel and are rendered by the same reader. A new interactive behavior is usually a new event type, not a new transport.

Choosing an OpenAI-flavored shape for the interior was pragmatic rather than ideological: most backends already speak it, so most adapters are nearly identity functions, and the `/v1` hub (Section 12) can re-emit the interior almost directly. The cost is that genuinely non-OpenAI providers (Anthropic's content-block model, Gemini's `parts`) do real translation work — but that work is confined to their adapter, which is exactly where the contract says it belongs.

5. The chat pipeline

`POST /api/chat` accepts a message, persists it, and returns a streamed NDJSON body: one JSON event per line, consumed on the client by a `ReadableStream` reader. NDJSON-over-POST was chosen over two obvious alternatives for concrete reasons. WebSockets bring a connection lifecycle — reconnection, heartbeats, state — that a request/response interaction does not need. Server-Sent Events cannot carry a POST body, which a chat turn requires. A streamed POST response is the smallest thing that works, and it composes cleanly with the interleaved event model of Section 4.

Two robustness properties are worth naming because they recur throughout the system. First, **messages are persisted as they complete**, so a conversation survives an application restart or a version upgrade, and the partial assistant text produced before a *provider* failure mid-stream is still written. A client that

disconnects is a different case — the generator is torn down rather than allowed to finish — and Section 6 describes the exit contract that governs it. Second, each conversation carries a **generation counter**. When history is rewritten out of band — a rewind, an edit-and-resend, a delete — the counter is bumped, and any still-running stream checks it before persisting. This is the mechanism that stops a "zombie" stream (client gone, cancellation not yet observed) from orphaning tool messages or resurrecting rows in a conversation the user already deleted. The same guard protects the research pipeline, which also persists into a conversation.

Treating the model as a plugin has a consequence for conversation state that is easy to miss: a workspace whose model is interchangeable will accumulate threads answered by *different* models. A conversation therefore records the provider and model it last used, and reopening one restores that pick. Without it, a global picker silently applies whatever model was last chosen anywhere to a thread every previous turn answered with another — a wrong-model reply that leaves no trace in the transcript. The same persistence instinct applies to the `usage` event: the token counts a provider reports at the end of a turn are stored on the assistant message they belong to, so per-reply accounting survives a reload instead of living only in the open tab. In agent mode, where one turn may make several provider calls, counts are recorded per step against the message that step produced.

Long conversations are trimmed before a turn begins: the system prompt and the newest messages always survive, the oldest turns fall off first, and a tool result whose originating tool call was trimmed away is dropped rather than sent as a dangling reference that confuses backends.

Two caveats belong here rather than in a footnote, because they are the sharpest edges in the current system. The budget is a fixed character count, identical for every model, not a token count derived from the one actually selected — the provider contract does not yet report a context window to size it against, so the same number is simultaneously conservative for a long-context model and optimistic for a small one. And in agent mode the trim runs once, before the first step; a tool-heavy turn appends results as it goes and can therefore grow past the budget within a single turn, which is precisely when it can least afford to. Section 19 treats both as open work rather than settled design.

6. Agent mode

Agent mode turns a conversation into a plan-act loop. The model receives the tool schemas alongside the conversation; any tool calls it returns are executed, their results appended as `tool` messages, and the loop repeats until the model stops calling tools or a step ceiling (fifteen) is reached. The step ceiling is a blunt instrument, but for a local tool it is the right kind of blunt: it guarantees termination without trying to be clever about what "stuck" means.

The builtin tools, and how each is gated, are the heart of the agent's safety posture:

Tool	Capability	Guard
shell	Run a PowerShell command	Per-call human approval
file_read / file_list	Read and list text files	Path sandbox
file_write	Write a text file	Free in workspace; approval outside it
web_search	DuckDuckGo search	—
web_fetch	Fetch a URL as readable text	Per-call approval ; refuses private IPs
memory_save / memory_delete / memory_search	Durable facts	Ungated, fully user-visible
knowledge_search	Retrieve indexed passages	Ungated, read-only
MCP server tools	Whatever the registered server exposes	Ungated — see Section 19

The approval gate. The one-way NDJSON stream cannot pause for a dialog box, so consent arrives out of band. When the agent proposes a gated call, the loop emits an `approval_required` event carrying a fresh approval id, then parks on an `asyncio.Future` registered under that id. A separate endpoint, `POST /api/approvals/{id}`, resolves the future when the user clicks Approve or Deny in the UI; the loop wakes and either runs the tool or reports "the user denied this tool call" as an ordinary tool result — so a denial redirects the model rather than crashing the run. Approvals have a timeout so an abandoned run cannot park forever, and they are single-shot: an unknown or already-used id is rejected. There is deliberately no "always allow."

The exit contract. A loop that can be cancelled at any await needs a defined answer for every way it can end, because the assistant message carrying a turn's

tool calls is written to the database *before* any tool runs. Both the OpenAI and Anthropic wire formats reject an assistant tool call with no answering tool result, so a run that dies between those two writes leaves a conversation that fails validation on every later turn — permanently, since the rows are already on disk. The loop therefore closes its own gaps. The tool phase is wrapped so that every exit path — a finished step, a provider error, the step ceiling, a denial, or a cancellation from Stop or a dropped connection — records a result for each call it announced, handing that write to a task outside the cancelled scope because a cancelled context cannot await anything. Results are recorded *before* they are streamed to the client, so a disconnect at the moment of announcement cannot discard work a tool genuinely completed. Every termination also persists a short note saying which ending it was, so a transcript never simply stops without explanation.

Reconstruction applies the same rule defensively rather than trusting the loop to have been the only writer: assembling history synthesizes a placeholder for any tool call it finds unanswered. That repairs conversations damaged before the contract existed, and any that a power loss or a hard kill still manages to interrupt. The two layers are deliberate — the loop handles the graceful cases precisely, and the repair pass covers the cases where nothing graceful was possible.

Per-call, argument-aware gating. Gating is not a static property of a tool but a decision made per invocation, through a `needs_approval(args)` hook. This is what lets `file_write` be frictionless inside the agent's own workspace yet require approval the moment its resolved target lands in a user-granted external folder — the same tool, two risk profiles, distinguished by the actual arguments.

The file sandbox. Relative paths resolve inside a dedicated workspace folder. Absolute paths are honored only inside folders the user has explicitly granted under Settings; every other path is refused with an error string the model can read and adapt to. Grants are data, surfaced to the model in its system prompt so it knows where it may work, and revocable at any time. Crucially, paths are re-resolved to their real location before the check, so a symlink or junction cannot be used to step outside a granted root.

MCP. Users can register stdio MCP servers by command line. Their tools are namespaced by server (`filesystem_read_file`) and merged into the agent's toolset, indistinguishable to the model from builtins. One implementation subtlety earned its place in the code comments: each server connection is owned by a single dedicated asyncio task, because anyio cancel scopes must be entered and exited by the same task; other tasks only use the already-initialized session object.

7. Persistent memory

An agent that forgets everything between conversations is a weaker assistant than the model's weights allow. Memory addresses this with deliberate modesty. The `memory_save` tool records a short, distilled fact; the newest memories, within a character budget, are injected into the agent's system prompt at the start of each agent-mode conversation, so preferences, project context, and decisions carry forward. `memory_search` reaches the older ones the budget omits, and `memory_delete` removes what has gone stale.

Memory is intentionally ungated, and the reasoning is worth stating because it runs opposite to the caution elsewhere. A saved memory has no effect outside the application; every save appears in the transcript as a tool card; and Settings → Agent memory is an always-available surface to review, add, and delete entries by hand. The combination — no external effect, full visibility, a standing kill switch — makes a per-save approval prompt pure friction. Plain, non-agent chat never receives memories at all, so the feature is scoped to the mode that opts into agency. Memories ride only on the request-local system prompt; they are never baked into a conversation's stored prompt, so deleting a memory truly forgets it everywhere.

8. Knowledge: local retrieval

Retrieval lets the workspace answer from documents far larger than any context window, without those documents ever leaving the machine — the local-first promise applied to the user's own corpus. Files or folders (text, code, PDFs) are indexed under Settings → Knowledge. Each source is extracted to text, chunked with a paragraph-aware sliding window that overlaps adjacent chunks so a fact split across a boundary is still retrievable, embedded through a provider that implements `embed()`, and stored as normalized `float32` vectors directly in SQLite alongside the chunk text.

The search itself is deliberately unsophisticated: embed the query, then compute a cosine similarity against every stored chunk and return the top matches. At the few-thousand-chunk scale this application targets, a brute-force dot product in pure Python completes in milliseconds, which buys a real architectural simplification — no vector database to run or bundle, no numpy dependency, and no index to keep consistent. The retrieval quality ceiling is lower than a dedicated

engine's, but for "chat with my folder of notes" it is comfortably sufficient, and the cost is a few dozen lines.

Two properties keep it safe and honest. Indexing is sandboxed to the same allowed roots as the file tools, re-resolving each walked file so a junction or symlink cannot pull outside-the-sandbox content into the index. And changing the configured embedding model clears the index, because vectors produced by different models are not comparable and silently mixing them would return plausible-looking nonsense. Retrieval surfaces to the agent as the read-only `knowledge_search` tool and is reused, as the next section describes, by deep research.

9. Deep research

Deep Research answers a question by reading the web and the local index, then writing a cited report. The notable design choice is that it is a **deterministic pipeline, not an autonomous agent loop**: plan, then gather, then read, then synthesize, as fixed stages. The reason is reliability with the smaller local models this application often runs. Asking a 7-billion-parameter model to correctly drive a fifteen-step tool loop toward a good report is a gamble; asking it to perform four well-scoped single-shot tasks is not. One completion turns the question into a handful of search queries; each query runs through web search and the candidate URLs are deduplicated; the top sources are fetched to readable text (under the same private-address protection as the agent's `web_fetch`) and the local knowledge index is consulted; a final streamed completion writes a Markdown report that cites its numbered sources, followed by a Sources list.

Because the fetched pages are untrusted text fed into a prompt, the synthesizer is hardened against injection: each source body is wrapped in explicit fenced delimiters and the system prompt states that fenced content is data to be summarized, never instructions to obey; source titles are sanitized before they enter the report so a crafted result cannot smuggle a Markdown image beacon or link. The whole run persists as an ordinary conversation, which means history, export, and the sidebar work with no new machinery — a recurring dividend of routing new features through the existing normalized model.

10. The writing editor

The editor is a local document workspace with AI editing folded in. Documents autosave to SQLite. A user selects text and applies a preset action (Improve, Shorten, Expand, Fix grammar), continues from the cursor, or issues a custom instruction; the model's suggestion streams into a panel to accept or reject. The edit endpoint, `POST /api/documents/edit`, is stateless — it returns only the replacement text and never mutates a document server-side. The client owns the decision to splice an accepted suggestion into the document at the captured selection, which keeps the server a pure text transformer and the document the single source of truth in the browser.

Two implementation details reflect hard-won correctness. Autosave is a dirty-flag debounce backed by a stale-proof mirror of the loaded document, and it *flushes* rather than merely cancels when the user switches documents or leaves the editor — so a fast edit-then-switch cannot silently drop the last change. And the text area is locked while a suggestion is pending, because an accepted suggestion is spliced at offsets captured when the run started; allowing edits in between would let those offsets drift and land the replacement in the wrong place. Both are the kind of bug that only appears under real use, and both were closed before the feature shipped.

11. The blind arena

The arena exists to answer a question the multi-provider core makes cheap to ask: which model is actually better for *my* prompts? It runs one prompt against two chosen models with their identities hidden — a coin flip decides which model fills which on-screen column, so position never leaks identity. Both answers stream in parallel through two calls to `/api/complete`, a stateless single-shot completion endpoint that the arena shares with any future feature needing a one-off generation. The user votes — A, B, tie, or both bad — which reveals the names and records the outcome to a local per-model win/loss/tie leaderboard. Almost all of the feature is a view; the routing that makes it possible was already built for chat.

12. The model cookbook

The cookbook is the one subsystem that reaches slightly outside the "model is a plugin" frame, and it does so carefully. It detects local hardware — CPU, RAM, and GPU VRAM, the last via `nvidia-smi` with a WMI fallback — using only the standard library and degrading every probe to "unknown" rather than raising, so a machine it does not fully understand still gets a usable page. It then rates a curated catalog of models as *fits your GPU / tight / CPU / too big* for that hardware, using rough per-model footprint estimates presented honestly as estimates. Models can be downloaded straight into Ollama with streamed pull progress and removed again.

The design keeps the plugin philosophy intact by being strictly *additive*: the cookbook never becomes the way models are served or selected. It only helps get weights onto disk via Ollama's native API, after which those models appear through the ordinary provider and the normal model picker. Downloading is therefore Ollama-specific — other backends get recommendations but not one-click installs — which is an honest limitation rather than a hidden coupling. Hardware detection, which shells out to external processes, runs off the event loop so a cookbook page load can never freeze a concurrent chat or research stream.

13. Interoperability

Syrudas is both a *client* of the OpenAI dialect, through `openai_compat`, and a *server* of it. The `/v1` surface exposes every model of every configured provider under a namespaced id (`ollama-local/llama3.1:8b`), and `chat/completions` translates each request into the normalized interior and routes it to the right backend, with streaming, tool calls, and usage preserved. These calls are stateless and never touch conversation history.

This inverts the usual integration burden. Rather than Syrudas writing an adapter for every editor and tool that might want to use it, any OpenAI-compatible client adopts Syrudas as its backend by changing one base URL — and in doing so gains uniform access to every model the user has configured, local and hosted alike, behind a single endpoint. The Continue extension for VS Code is the reference consumer, and gains something a chat window cannot offer — inline completions and edits applied directly in the editor. Command-line tools such as `aider` work the same way.

An earlier release also shipped a dedicated VS Code extension that embedded the workspace UI in an editor panel. It was removed: it wrapped the same web app in

an iframe rather than integrating with the editor, so it duplicated what a browser tab already did while adding a separate distribution channel and a version-drift surface. The one capability worth keeping — launching a chat prefilled from an external caller — was never the extension's to begin with. It is a `?prompt=` parameter honored by the web app itself, and remains available to any caller.

14. File attachments

Attachments follow a stateless pipeline that mirrors the rest of the system's preference for keeping state in one place. The client uploads a file to `POST /api/attachments`; the server extracts text — UTF-8 for text-like formats, pypdf for PDFs, binaries rejected, with a character cap and explicit truncation flags — and returns it. The client then embeds that text into the outgoing message inside `<file name="...">` delimiters.

Storing attachment content *inside the message* rather than in a side table of blobs buys three properties at once: replaying history to any provider needs no special casing, because the file is just message text; backup and export are simply the database file; and the UI reconstructs collapsible per-file chips from the message text alone. The trade-off is larger message rows, which is acceptable at the capped sizes. For documents too large for the model's context window, the knowledge subsystem (Section 8) is the better path, and the two features compose: attach for a one-off, index for a corpus.

15. Security and privacy model

The threat model is honest about what a local, single-user tool can and cannot promise, and the posture is layered.

- **Network exposure.** The server binds to `127.0.0.1` exclusively, and a middleware rejects any request whose `Host` header is not a loopback name. Binding alone is not enough: without the Host check, a web page the user is merely visiting could POST to the local server via DNS rebinding and drive it from the outside. The two together close that vector. With no remote surface, there is deliberately no authentication layer to misconfigure.

- **Server-side request forgery.** The web-fetch path — used by both the agent

and deep research — refuses any URL that resolves to a private, loopback, or link-local address, and re-checks on every redirect hop, so a fetched page cannot bounce the agent onto localhost, the application's own API, or the local network.

- **Model-initiated actions.** The genuinely dangerous capabilities — arbitrary shell, web fetch, and file writes outside the workspace — are each gated per call with no persistent allow. File access is deny-by-default outside the workspace plus explicit grants. Instruction/data separation is currently uneven, and is stated here rather than glossed: the deep-research pipeline fences untrusted source text in explicit begin/end markers, strips counterfeit markers a page may itself contain, and instructs the model to treat the contents as information even where the text claims otherwise. Agent mode does not yet do this — tool output, including fetched pages and retrieved passages, is appended to the conversation unmarked. Giving the agent loop the same fencing is tracked work, not a design position.

- **Static file serving.** The single-page-application fallback route resolves request paths against the bundled web assets. Constraining that resolution so no request path can escape the asset directory is an outstanding fix; until it lands, the local HTTP surface should be assumed capable of reading files the user can read, and anything with access to the port should be treated accordingly. It is the first item on the current work list.

- **Data at rest.** Conversations, settings, memories, indexed chunks, documents, and provider API keys live in one local SQLite file. Keys are stored in plaintext — an OS-keychain integration is future work — but are masked (•••1234) in every API response, so the browser never re-receives a full secret. The honest consequence, stated plainly in the setup guide, is that anyone with read access to the data folder can read the keys; the folder should be treated as being as sensitive as the keys themselves.

- **Telemetry.** None. The application makes outbound connections only to the model backends the user configured and to URLs the user, or the user's agent under the rules above, requests.

- **Residual risks.** A granted folder is fully readable and writable by any model the user runs, including a poorly aligned local one; MCP servers run with the user's privileges and are trusted by the act of registration; and the unsigned executable requires a one-time SmartScreen click-through. These are documented

rather than hidden, because a local tool's security story is only as good as its honesty about the edges.

16. Accessibility and theming

Appearance and colour vision are treated as two independent axes, so a colour-blind user is never forced to choose between an accessible palette and a preferred background. Appearance (system, light, or dark) and colour vision (default, protanopia, deuteranopia, tritanopia, achromatopsia) are applied as attributes on the document root and resolved entirely through CSS variable overrides, with a pre-render inline script setting them before first paint to avoid a flash of the wrong theme. Colour-vision modes remap only the four status hues and are tuned per background so status *text* clears WCAG AA contrast on both light and dark; achromatopsia drops hue entirely and relies on luminance. Underpinning all of it is a rule the UI follows regardless of palette: status is never conveyed by colour alone — every state also carries an icon and a text label — so meaning survives any colour transformation, and the palettes are a legibility improvement rather than a load-bearing signal. The application also honours the operating system's reduced-motion preference.

Accessibility extends past colour. Every interactive control carries an accessible name — icon-only buttons included, each labelled with the action and its target (for example, the specific conversation a delete button removes) — and the conversation and document lists, which are visually rows, are exposed as keyboard-focusable buttons with the active row marked. Actions that appear on hover — the per-message copy and edit controls, the per-code-block copy button — are faded rather than hidden outright, because a control hidden with `visibility` is removed from the tab order entirely and can never receive the focus that would reveal it; kept at zero opacity they remain focusable and appear on keyboard focus.

The honest boundary of that work is keyboard operability, not screen-reader support, and the distinction matters enough to draw. Named controls and a correct tab order are necessary for a screen reader and not sufficient for one: the application currently declares no live region, so streamed replies and tool state changes are not announced as they arrive, and the collapsible tool card remains the one control driven by pointer events alone. A user navigating by keyboard is well served today; a user listening is not yet, and the gap is specific enough to close — a live region on the transcript, message-level status announcements

rather than per-token ones, and keyboard semantics on that card. That work, and a screen-reader pass to validate it, is Section 19 material.

17. Testing and assurance

A one-person codebase of this breadth stays trustworthy only with a testing discipline that matches. Every subsystem ships with an offline test suite that drives the *real* code paths against fakes rather than mocks of its own logic: a scripted fake provider exercises the agent and research loops through their actual event handling; `httpx.MockTransport` stands in for the Anthropic and Gemini wire protocols and for Ollama's native API; a deterministic fake embedder makes retrieval ranking meaningful without a model. Because the fakes sit only at the true boundaries — the network, the model — the suites verify the system's own behavior, and they need no network, no GPU, and no running model, so they run anywhere in seconds.

Cancellation is tested on the same principle, because it is the behavior least likely to be exercised by hand and the most expensive to get wrong. A dedicated suite drives the real loop, interrupts it mid-tool-call and mid-approval, and asserts what must hold afterwards: no tool call left unanswered, no approval id leaked, a recorded reason for the ending, a completed result never replaced by a placeholder, and a history that still assembles into a valid provider request. Fakes make those assertions cheap and deterministic; because a scripted provider cannot reproduce the fragmented, interleaved tool calls a real backend streams, the same properties were also confirmed against a live local model.

That property is what makes the suites enforceable rather than merely available. A single entry point runs every offline suite alongside the frontend's unit tests, lint and typecheck, and continuous integration invokes that same entry point on every push and pull request, on Windows because that is the platform the application targets. The value is less in the automation than in the guarantee it removes from human memory: a regression in any subsystem fails the branch that introduced it, rather than waiting for whoever next thinks to run the suite by hand.

The client is covered on the same principle. Adversarial review kept finding its defects in one place — the logic that folds a stream of events into a thread and rebuilds that thread from storage — because those are stateful transforms whose mistakes are invisible until a specific sequence occurs. That logic is therefore kept as pure functions outside the component: a reducer that folds one event into the

item list, and a rebuild that turns saved messages back into it. Both are exercised directly, with event scripts standing in for a model, including the sequences that previously produced wrong output — an agent step that calls a tool without speaking, where the turn's token counts belong to no visible message and must be discarded rather than written onto the previous answer. Interaction that genuinely depends on the DOM, such as the sidebar's rename committing on Enter and blur but discarding on Escape, is driven through the rendered component instead.

Beyond the suites, substantial features were put through an adversarial review before shipping: independent passes over the diff for correctness, security, frontend contract, and test coverage, with each finding challenged by skeptics before it was accepted, and every confirmed finding fixed and re-verified. Several of the correctness properties described earlier — the autosave flush-on-switch, the editor's offset lock, the research injection hardening, the cookbook's off-loop detection — are fixes that this process surfaced. The point is not that the code is flawless but that its most load-bearing behaviors have been attacked on purpose.

18. Packaging and distribution

The release artifact is a portable zip: one PyInstaller onefile executable (windowed, over WebView2), the optional provider connectors, an end-user README, the setup guide, and the MIT license. Everything mutable lives beside the executable, so an installation is trivially copyable, movable, and deletable. On first run with an empty database, the server probes the well-known local backends — Ollama on `:11434`, LM Studio on `:1234` — and auto-configures any that respond, so a user who already runs a local model server reaches a working chat with zero configuration.

Version identity flows from `APP_VERSION` into the health endpoint, the UI footer, and — as of this revision — the cover of this document, substituted at render time rather than typed. One duplicate remains, and it is worth naming because the previous edition of this paper claimed the opposite: the Windows file properties of the executable are read from a separate `version_info.txt` that is maintained by hand, so cutting a release edits two files rather than one. Until that file is generated from `APP_VERSION` as well, the two can drift — and a shipped PDF whose cover disagreed with the build it accompanied is exactly what that drift looks like. Two Windows-specific build lessons are preserved in the codebase for posterity: PyInstaller bundles are invisible to `pkgutil`, which is why builtin provider adapters are imported statically; and PowerShell 5.1 under `$ErrorActionPreference = "Stop"`

turns a native command's harmless stderr into a terminating error, which is why the build scripts wrap native calls in `cmd /c "... 2>&1"`.

19. Limitations and roadmap

The honest limitations are the mirror image of the design choices. This edition states the engineering ones alongside the product ones, because the previous edition listed only the latter — and the engineering ones are what a reader evaluating the system would actually want to know.

Product boundaries. The application is single-user by intent and Windows-only in a deeper sense than packaging: the shell tool spawns PowerShell directly, hardware detection reads WMI, the desktop shell targets WebView2, and the build and run scripts are PowerShell throughout. The web layer is portable Python, but a port would be genuine work rather than a packaging change, and everything below is written on the assumption that Windows remains the target. Multimodality is text-only: normalized message content is a string, a deliberate simplification that also means a vision-capable model can be selected but not fed an image — a mismatch the bundled model catalogue should not advertise until it is resolved in one direction or the other. Key storage is plaintext on disk. Model *download* management is specific to Ollama.

Runtime limitations. These matter more day to day than the boundaries above. The context budget is a fixed character count rather than a model-aware token count, and in agent mode it is applied once per turn rather than once per step (Section 5), so a tool-heavy run can evict the user's original request while it is still working on it. The file tools offer read, whole-file write, and list — no patch-style edit, no pagination, no search — so changing part of a large file means rewriting all of it, which is wasteful and, when a read was truncated, a data-loss shape rather than merely an inefficient one. Tool arguments are not validated against the schema the tool declares before dispatch. MCP tools inherit no approval gate, which makes registering a server the one action that grants the model more latitude than the builtin toolset does. And agent mode still lacks the instruction/data fencing that deep research already applies (Section 15).

Instrumentation. A run leaves a transcript but not a record: no run identifier, no step timings, no aggregated cost, no behavioral evaluation suite. The practical consequence is that the two highest-leverage knobs in the system — the agent's system prompt and its tool descriptions — can be changed without any signal that

behavior moved. Run identity and a small fixed task set to measure against are therefore near the top of the list, because they are the prerequisite for improving anything else with confidence rather than by impression.

Structural. Agent mode and deep research are two orchestration loops that share a persistence helper and little else, so a fix applied to one is not a fix to the other. The tool interface accepts arguments but no run context, which is the pinch point for several items above. Both are the kind of change that grows more expensive with every feature layered on top, which is an argument for doing them earlier than their urgency alone would suggest.

Direction. In rough order of value: the runtime limitations above; run instrumentation and evaluations; image input carried through the same normalized schema; OS-keychain key storage; and an optional local token on the /v1 hub for shared machines. Productivity integrations — email, calendar, notes — remain the largest gap relative to a full personal-assistant suite, and each is effectively its own subsystem that would be evaluated on its own terms rather than folded in for completeness.

20. Conclusion

Syrudas AI is an argument, made in code, that a genuinely capable AI workspace — streaming chat, consent-gated agency, local retrieval, deep research, model comparison, a writing editor, protocol interoperability, and one-click distribution — fits inside a small, single-maintainer codebase when one abstraction is chosen well and enforced everywhere. Each feature in this paper is less an invention than a consequence: route it through the normalized model and the provider contract, and history, export, streaming, and provider-independence come for free. Models will keep changing, faster than anything else in the stack. A workspace whose only firm opinion about models is a five-method contract is built to outlast every one of them.

Syrudas AI is open source under the MIT license. Architecture reference and setup: `README.md` and `docs/SETUP.md` in the repository at github.com/LenSyrudas/syrudas.